5-Mark Questions (Q1 & Q2) - Expect:

Type 1: Algorithm Implementation

- "Draw a line from (1,1) to (8,5) using **Bresenham's algorithm**"
- "Use **DDA algorithm** to rasterize line from (2,3) to (9,7)"
- "Apply **Midpoint Circle algorithm** for circle with center (5,5) and radius 4"

Type 2: Short Numerical

- "Apply transformations: Translate (3,4), then Rotate 90°, then Scale (2,3) to point P(2,1)"
- "Find transformation matrix for rotating about point (3,2) by 45°"

Type 3: Conceptual Comparison

- "Compare **DDA vs Bresenham** line algorithm (3+2 marks)"
- "Differentiate between **Raster and Vector graphics**"
- "Explain **World view vs Screen view** transformation"

10-Mark Question (Q3) - Expect:

Type 1: Complete Algorithm with Steps

- "Explain **Bresenham's Circle algorithm** with complete derivation and draw circle with center (0,0), radius 5" (4 marks theory + 6 marks numerical)

Type 2: Transformation Problems

- "Derive **2D composite transformation matrices**. Apply sequence: Scale(2,2) → Rotate(45°) → Translate(5,3) to triangle with vertices A(1,1), B(3,1), C(2,3)" (4 marks derivation + 6 marks calculation)

Type 3: Comprehensive Theory

- "Explain the complete **graphics pipeline** from object space to screen space with coordinate transformations" (10 marks theory)
- "Discuss **color models (RGB, CMYK, HSV)** with conversion formulas and applications"

Time Management Strategy

Recommended Approach:

Minutes 0-5: Quick scan of all questions, decide order

- Start with your **strongest topic** for confidence
- Solve easier questions first

Minutes 5-20: First 5-mark question (15 min max)

Minutes 20-35: Second 5-mark question (15 min max)

Minutes 35-55: 10-mark question (20-25 min)

- If it's numerical: Show ALL steps
- If it's theory: Use diagrams, bullet points

Minutes 55-60: Review, check calculations

Quick Revision Priorities

Must Know (High Priority):

1. **Bresenham Line Algorithm**
    - Formula: d = 2dy - dx
    - Step-by-step process
2. **DDA Algorithm**
    - x_new = x_old + 1
    - y_new = y_old + m
3. **Transformation Matrices**
4. Translation: [1 0 tx]
5. [0 1 ty]
6. [0 0 1]
7.
8. Rotation: [cosθ -sinθ 0]
9. [sinθ cosθ 0]
10. [0 0 1]
11.
12. Scaling: [sx 0 0]
13. [0 sy 0]
14. [0 0 1]
15. **Midpoint Circle Algorithm**
    - Decision parameter: P_k
    - 8-way symmetry points

Writing Tips for Maximum Marks

For 5-Mark Questions:

- Write 5-7 key points
- Include **one small diagram** if relevant
- Show calculation steps clearly
- Underline final answers

For 10-Mark Questions:

- Introduction (1 mark)
- Main content with subheadings (7-8 marks)
- Diagram/numerical work (neat!)
- Conclusion (1 mark)

Golden Rules:

✅ **Always write formulas** first ✅ **Show all steps** in numerical problems ✅ **Draw neat diagrams** (use ruler if available) ✅ **Label everything** clearly ✅ **Circle/box final answers** ✅ **Use proper terminology** from the topic list

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5-MARK QUESTIONS

Q1. Compare Raster Graphics and Vector Graphics (5 marks)

Answer:

Aspect	Raster Graphics	Vector Graphics
Definition	Composed of pixels in a grid	Composed of mathematical paths/objects
Scalability	Loses quality when scaled up (pixelation)	Infinitely scalable without quality loss
File Size	Large file sizes for complex images	Smaller file sizes
Best For	Photographs, complex images, realistic art	Logos, illustrations, text, diagrams
Formats	JPG, PNG, GIF, BMP, TIFF	SVG, AI, EPS, PDF
Editing	Pixel-by-pixel editing	Object-based editing
Examples	Digital photos, scanned images	Company logos, fonts, technical drawings

Memory Calculation:

- Raster: width × height × color_depth
- Vector: Stores only object definitions (points, curves, colors)

Q2. Explain RGB Color Model with its representation (5 marks)

Answer:

RGB Color Model (Additive Color Model):

RGB stands for Red, Green, and Blue - the primary colors of light.

Characteristics:

- **Type:** Additive color mixing
- **Used in:** Computer monitors, TVs, digital cameras, scanners
- **Range:** Each component ranges from 0-255 (8 bits per channel)
- **Total Colors:** 256 × 256 × 256 = 16,777,216 colors (24-bit True Color)

Color Representation:

```
Black:   RGB(0, 0, 0)
White:   RGB(255, 255, 255)
Red:     RGB(255, 0, 0)
Green:   RGB(0, 255, 0)
Blue:    RGB(0, 0, 255)
Yellow:  RGB(255, 255, 0)
Cyan:    RGB(0, 255, 255)
Magenta: RGB(255, 0, 255)
```

Q3. What is Color Depth? Explain different color depths (5 marks)

Answer:

Color Depth (Bit Depth): Number of bits used to represent the color of a single pixel.

Different Color Depths:

1. **1-bit (Monochrome)**
    - Colors: 2^1 = 2 colors (Black & White)
    - Memory: 1 bit per pixel
    - Use: Simple graphics, text
2. **4-bit (16 colors)**
    - Colors: 2^4 = 16 colors
    - Memory: 0.5 bytes per pixel
    - Use: Old CGA graphics
3. **8-bit (256 colors)**
    - Colors: 2^8 = 256 colors
    - Memory: 1 byte per pixel
    - Use: GIF images, indexed color

4. **16-bit (High Color)**
    - Colors: 2^16 = 65,536 colors
    - Memory: 2 bytes per pixel
    - Distribution: 5 bits Red, 6 bits Green, 5 bits Blue
5. **24-bit (True Color)**
    - Colors: 2^24 = 16,777,216 colors
    - Memory: 3 bytes per pixel (8-8-8 RGB)
    - Use: Standard for digital images
6. **32-bit (True Color + Alpha)**
    - Colors: 24-bit color + 8-bit alpha channel
    - Memory: 4 bytes per pixel
    - Alpha: Transparency information

Memory Calculation Example: For 1024×768 image with 24-bit color: Memory = 1024 × 768 × 3 bytes = 2,359,296 bytes ≈ 2.25 MB

Q4. Differentiate between CMYK and RGB color models (5 marks)

Answer:

Parameter	RGB	CMYK
Full Form	Red, Green, Blue	Cyan, Magenta, Yellow, Black (Key)
Type	Additive	Subtractive
Primary Colors	Red, Green, Blue	Cyan, Magenta, Yellow
Color Mixing	Adding colors creates white	Adding colors creates black
Used In	Screens, monitors, digital displays	Printing, printers, physical media
Color Range	0-255 for each channel	0-100% for each channel
White Color	(255, 255, 255) - All on	(0, 0, 0, 0) - No ink
Black Color	(0, 0, 0) - All off	(0, 0, 0, 100) - Full black ink
Application	Web design, digital photography	Magazines, brochures, books

Why CMYK has 'K' (Black)?

- Theoretically C+M+Y should give black
- In practice, gives muddy brown
- Adding pure black (K) gives true black and reduces ink cost

Conversion consideration: RGB has wider gamut than CMYK - some RGB colors cannot be printed accurately.

Q5. Explain pixel representation and frame buffer organization (5 marks)

Answer:

Pixel (Picture Element):

- Smallest addressable element of an image
- Contains color information
- Position identified by (x, y) coordinates

Pixel Representation:

1. **Binary Images:** 1 bit per pixel (0 or 1)
2. **Grayscale:** 8 bits per pixel (0-255 intensity)
3. **Color Images:** 24 bits per pixel (8R + 8G + 8B)

Frame Buffer: A dedicated memory area that stores pixel values for display.

Organization:

```
Screen Resolution: W × H pixels
Color Depth: D bits per pixel
Frame Buffer Size: W × H × D bits
```

Example: 1920×1080 display with 24-bit color

- Buffer size = 1920 × 1080 × 24 bits = 49,766,400 bits ≈ 6 MB

Memory Mapping:

- **Linear mapping:** Pixels stored row by row
- **Packed pixel format:** RGB values stored consecutively
- **Planar format:** Separate planes for R, G, B

Refresh Process:

- Video controller reads frame buffer 60+ times/second
- Converts digital values to analog signals (or digital for LCD)
- Updates display continuously

Double Buffering:

- Two frame buffers: front buffer (displayed), back buffer (being drawn)
- Prevents flickering and tearing